

Installation and Operation Manual

IXS Series X-Ray Generator



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VJ X-Ray (NY) LLC
95 Carlough Road
Bohemia, NY 11716 USA
Phone: (631) 589-8800
Fax: (631) 218-1531
Email: info@vjt.com

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1.0 Introduction

This operational manual is intended to assist the user work with our products safely. The entire manual must be read thoroughly and carefully. It contains the necessary instructions to safely operate our integrated x-ray sources (IXS). All products must be operated in accordance with these safety instructions and your system must adhere to local statutory regulations.

The IXS series is an exceptionally compact, lightweight, and reliable line of integrated X-ray sources. The X-ray source integrates a high voltage inverter, a filament supply, an X-ray tube, a shielded X-ray housing, with a heat exchanger (if needed) into one unit. The product is designed to comply with the Safety Requirement for Electrical Equipment for Measurement, Control and Laboratory Use (UL61010-1) and in compliance with 21CFR1020.40 Cabinet X-ray System Requirements. Main features of the product include:

- Oil based insulation provides efficient thermal dissipation and better thermal stress management.
- Integrated closed loop cooling for optimization of output power and product reliability.
- The source can be controlled via a Serial or an Ethernet connection.

1.1 General Safety information

- Safety regulations must always be adhered to.
- Electrical Safety - Ground conductors must be installed in all components in accordance with the safety standards and requirements. Ground conductor junctions must always be in good condition.
- Radiation Protection – It is the product owner's responsibility to observe and comply with the legal and regulatory practices pertinent in the operating environment when installing and using the equipment. Necessary registrations, authorizations and approvals must be obtained before using the X-Ray equipment. Safety measures and proper shielding practices must be taken to prevent person being exposed to radiation dose that is higher than permitted by regulations.
- Material Handling and Disposal - High voltage power supplies and X-Ray tubes contain materials that may endanger the environment, and thus must be handled or disposed in an environmental friendly way. Oil, greases, lead parts, and beryllium may not be disposed together with normal industrial or domestic waste. Local regulation for disposal must be observed.

1.2 Unpacking

Check the boxes against the packing slip to ensure all components are present.

Remove the components from their shipping container. Do not discard the packing materials; retain the packing materials and carton in case the unit needs to be returned. Inspect the unit for dents or other damage, which may have occurred during shipping. In the unusual event that shipment damage has occurred, contact the transportation company and the manufacturer immediately.

If additional materials have been enclosed in the shipping crate (cables, coolers etc.), inspect them as well. Then put them aside and DO NOT proceed until this operation manual has been completely read and understood.

2.0 IXS Setup

This chapter describes all connections and detail interfaces. Depends on what IXS model you had purchased. The appearance of control box panel might vary slightly, but the connector notation should be the same.

2.1 Connections Overview

Fig 2-1 shows an overview of how the Integrated X-ray Source (IXS) is connected. Fig 2-2 to 2.5 details various control box panel.

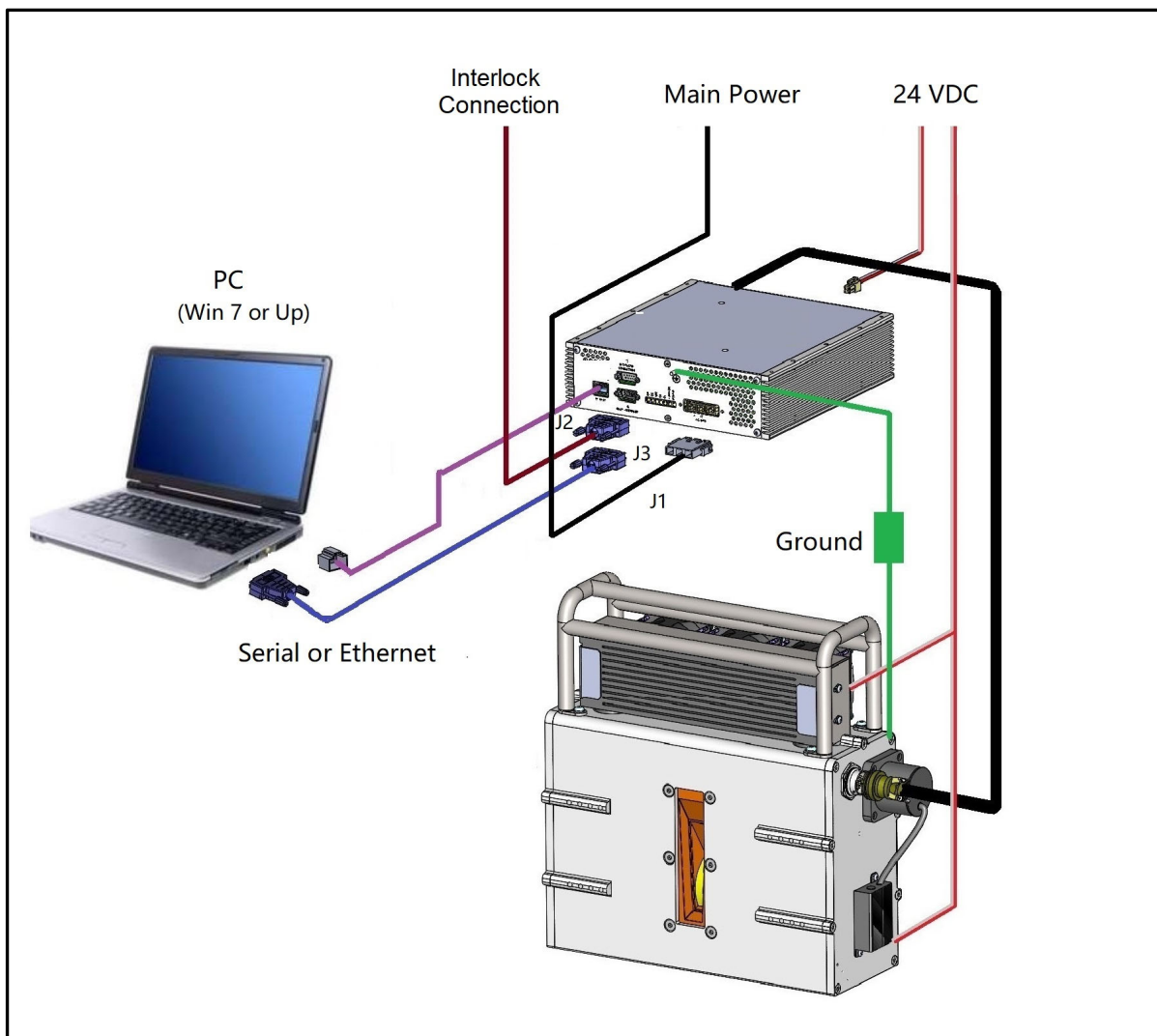


Fig 2-1 IXS Connection Diagram

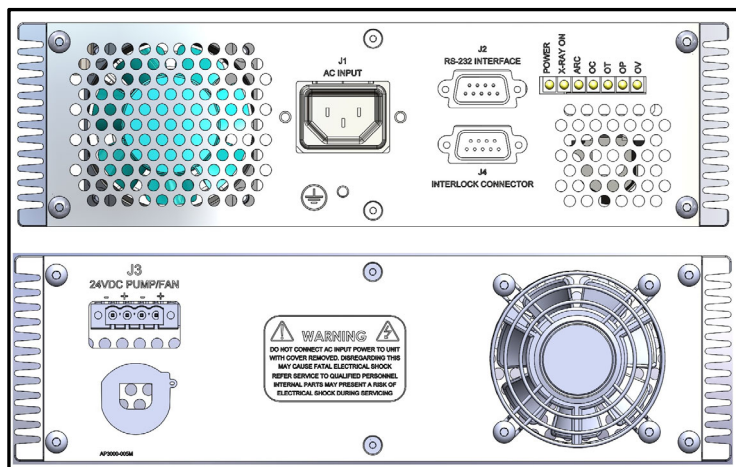


Fig 2-2 Type A Control Box

Connector	Description
J1	Main Input Power
J2	Serial (RS232) Interface
J3	24VDC Pump/Fan Output
J4	Interlock Connection

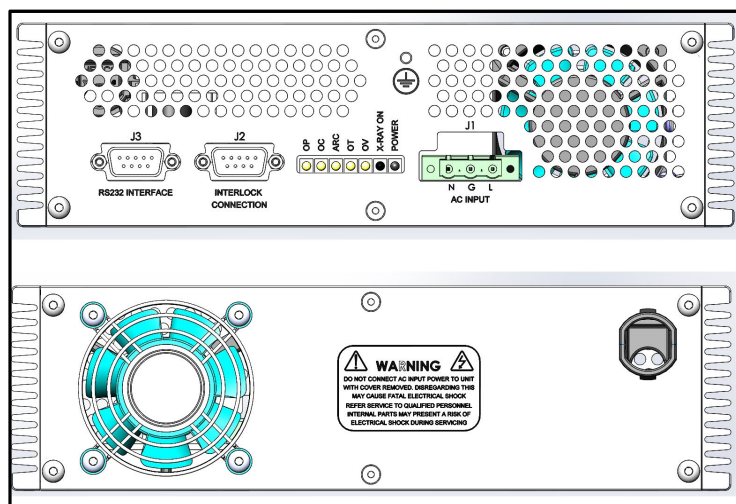


Fig 2-3 Type B Control Box

Connector	Description
J1	Main Input Power
J2	Interlock Connection
J3	Serial (RS232) Interface

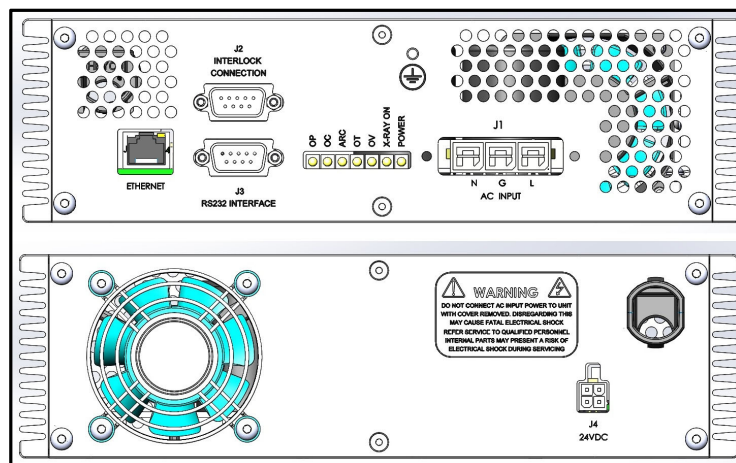


Fig 2-4 Type C Control Box

Connector	Description
J1	Main Input Power
J2	Interlock Connection
J3	Serial (RS232) Interface
J4	24VDC Customer Input
Ethernet	Ethernet (RJ45) Digital Interface

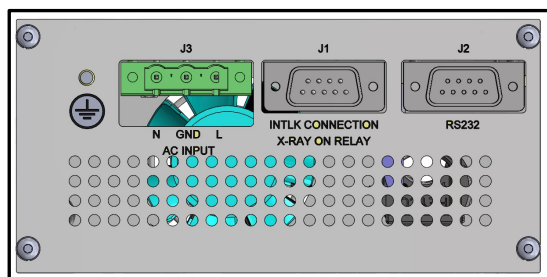


Fig 2-5 100kV Control Box

Connector	Description
J1	Interlock Connection
J2	Serial (RS232) Interface
J3	Main Input Power

2.2 Main Input Power

This is the main AC Input. This AC input power must be connected to the Safety Interlock circuits of the X-Ray Cabinet such that door opening results in physical disconnection of this input power (in compliance with 21CFR1020.40 Cabinet X-Ray Systems requirements).

Depending on your specific control box, we have three type of AC connectors. The polarity marking (L, N and G) are stenciled on the panel. The input power cable or connector is shipped with the IXS unit. It needs to be properly terminated to the facility panel.

- Type A Control Box: IEC 320 AC Plug
- Type B/100kV Control Box: Phoenix 1860232 Terminal Block
- Type C Control Box: AMP 556882-3 Terminal Block.

It is highly recommended to install an isolation transformer between the facility panel and IXS's AC input. This is to provide isolation and suppress electrical noise. We recommend the transformer be 50% above rated current. For example: a 3 Amp IXS: $3(A) \times 230 (VAC) \times 1.5 = 1 \text{ kW}$

Input Power: 230VAC $\pm$ 10%, 50/60Hz, Single Phase. 15A

2.3 Interlock Connection

This is the interlock connection; it uses a 9 pin male connector. The unit ships only with the connector, the customer is responsible to make the corresponding cable. Refer to the table below for the pin out.

Pin Out	Name
1	Interlock Out (+24VDC)
2	Interlock In (+24VDC)
3	X-Ray On Relay contact Common
4	X-Ray On Relay contact N/C
5	X-Ray On Relay contact N/O
6	Pre-warning Relay Contact Common (If equipped)
7	Pre-warning Relay Contact N/O (if equipped)
All other pins	Not Used

Pin 1 & 2 Safety Interlock:

This Safety Interlock contains two pins, Interlock Out, J2-1 (+24V) and Interlock In, J2-2. These two pins are intended for external Safety interlock switches of the cabinet system. X-Rays will not be produced & interlock open message will be displayed unless J2-1 and J2-2 are connected through safety switches. When J2-1 and J2-2 are not connected, the power of the High Voltage transformer driver will be cut off through a solid-state relay. (Fig 2-3 is a Typical External Circuit diagram)

Pin 6 & 7 X-Ray Pre-Warning (If equipped):

The X-Ray Pre-warning Relay (SSR) J2-6 and J2-7 *will close to indicate the X-Ray output is ON*. The Pre-Warning timer will start when the X-Ray Enable button is pressed and will determine the amount of time until the X-Ray is actually on. This timer is programmable between 0 to 60 seconds.

J2-6 and J2-7 is configured as a normally open circuit. An external 24V, 1Amp max can be connected a Pre-Warning device such as a Blinking LED or Alarm to J2-7. J2-6 should be connected to Ground.

Pin 3, 4 & 5 X-Ray On Relay (Solid-State Relay - SSR):

The X-Ray On Relay will Close (J2-3 and J2-5) when HV is enabled and there are no faults occurred.

The X-Ray On Relay will Open (J2-3 and J2-5) when the X-Ray output is OFF.

J2-3 and J2-5 are configured as a normally open circuit. An external 24V, 100mA-200mA load can be connected to J2-3 to drive an "X-Ray On" device such as a LED or Lamp (Fig 2-6).

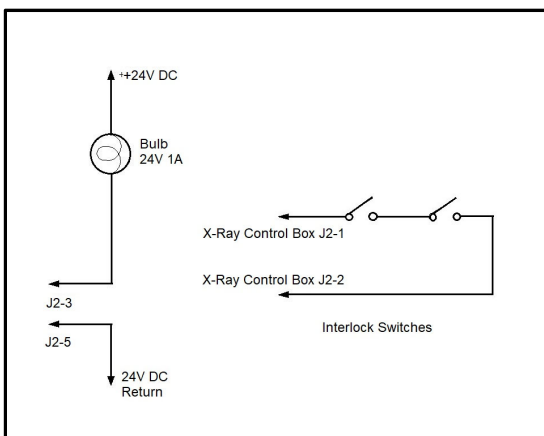


Fig 2-6 External Safety Circuit Diagram

2.4 Serial (RS232) Interface

The IXS is controlled through a computer via a Serial (also call RS232) or an Ethernet interface. Below is the pin out of Serial connector (9 Pin sub-D Female). For software control, please reference to Chapter 3.

NOTE: When both Serial and Ethernet (if available) are connected, Ethernet is the default interface. The Serial interface will be automatically disabled.

Pin Out	Name
2	TX- (Transmit)
3	RX+ (Receive)
5	Signal Ground
All other pins	Not Used

2.5 Ethernet Digital Interface (if equipped)

The Ethernet interface is an alternative to the Serial connection. Both interfaces are mutually exclusive; you can only use one at a time with the Ethernet connection being the primary. The interface supports the following attributes:

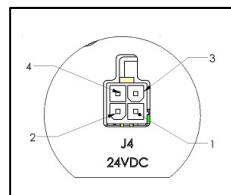
- 10-Base-T (rev A-B), 10/100-Base-T (rev C and higher)
- The IP address, Network Mask and TCP port number can be set using a special program (see chapter 5.3).
- Network attachment via Crossover and Standard Ethernet cables.
- Supported Operating System: Windows 7 or higher.

Pin Out	Name
1	TX+
2	TX-
3	RX+
4-5	Not Used
6	RX-
7-8	Ground

2.6 Housekeeping 24 VDC

This housekeeping 24VDC (Molex 39-30-0040) is supplied by customer. It is used to maintain control circuit and pump operation in event of IXS power failure for proper system shutdown.

Pin Out	Name
1	+24VDC@3.0A for Pump
2	-24VDC Return for Pump
3	+24VDC@1.5A for Control Circuit & Fan
4	-24VDC Return for Control Circuit & Fan



2.7 System Ground

It is vital for IXS to be grounded in order to function properly and safely. Connect a facility ground Wire (12 AWG minimum) to the IXS's control box ground terminal and on the IXS tank unit.

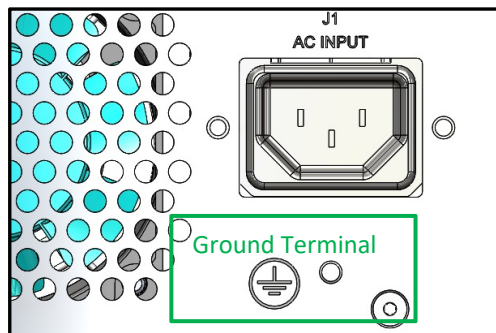


Fig 2-7 System Ground

2.8 Cooler Connection

If the x-ray source is configured with an external cooler, please refer to the VJ X-Ray General Purpose Cooler Operation and Maintenance Manual or the relevant operational manual from the chiller or cooler manufacturer.

3.0 Software Operation

This section provides instructions on setting up the Graphic User Interface (GUI) of IXS. The GUI requires a Win XP (or up) PC with either Ethernet or Serial connected to the IXS.

3.1 GUI setup

Make sure all connections are tightly secured, ground wires are properly installed before turning on power. Follow below steps to run the control software:

1. Insert CD to the system computer. Double click the **setup.exe** to install the GUI (Fig 3.1). The GUI installer will open as shown in Fig 3-2.

Note: If you have **Windows 10 or up**, you need to run the **setup.exe** program in Administrator mode, see section 5.2



Fig 3-1 Installer

2. After the installer opens click **Next** for each of the three windows as shown in Fig 3-2, Fig 3-3, and Fig 3-4. The installer will then install the GUI. When done click on **Finish** as shown in Fig 3-5 and restart the PC.



Fig3-2 Welcome

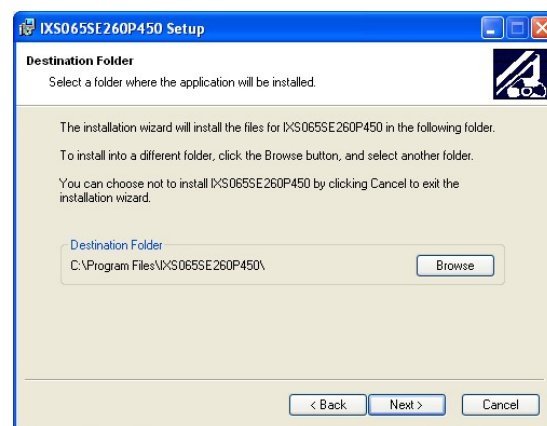


Fig 3-3 Program Location

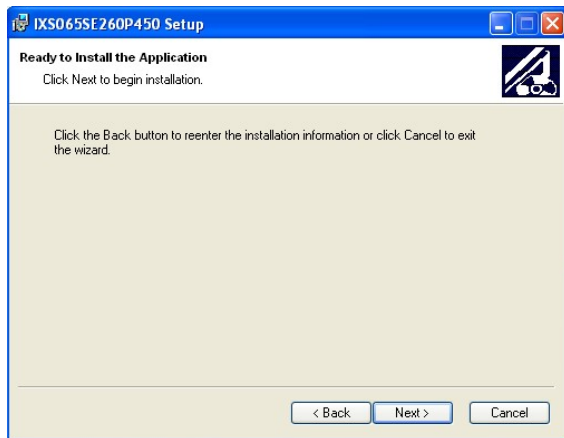


Fig 3-4 Install

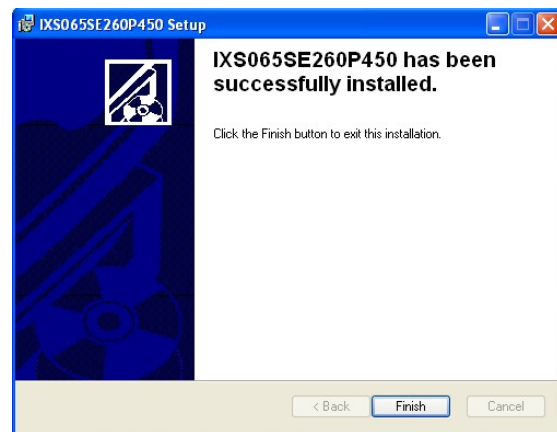


Fig 3-5 Finish

3. Once the PC is restarted, go to **VJ X-Ray** or **VJ Technologies** in the Start menu and click on the name of your model number to open the GUI. (Fig 3-6)
Note: If you have **Windows 10 or up**, you need to run the program in Administrator mode, see section 5.2

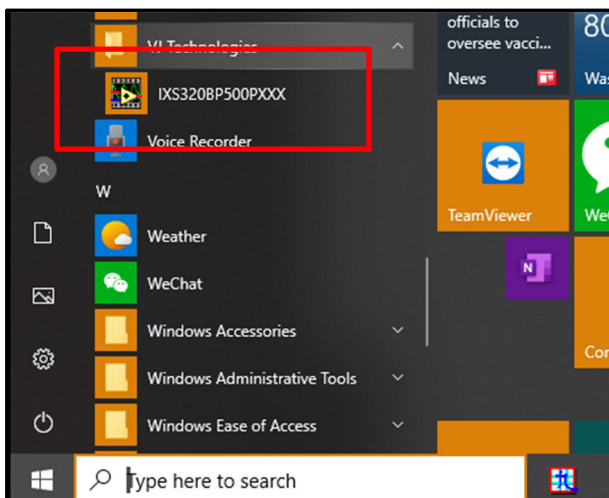


Fig 3-6 IXS run Program

4. If the IXS is equip with an ethernet port, then the GUI will open with a pop-up window that asks you to choose ethernet or serial connect as shown in Fig 3-7.
5. If the IXS has no ethernet port or you select serial, then go to step 6 otherwise go to step 7.

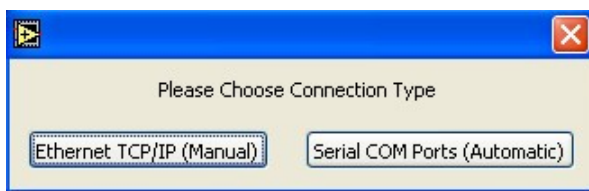


Fig 3-7 Connection Type Select Pop-up Window

6. If you choose to connect the ethernet port, then the GUI will open as shown in Fig 3-8. Make sure the IP address is 192.168.10.120, then press Connect. When successfully connected it will display the IP address used in the upper left-hand corner just below **Initializing Software...**

Note: If you need to change the Ethernet's IP address, please see instructions on Chapter 5.3.

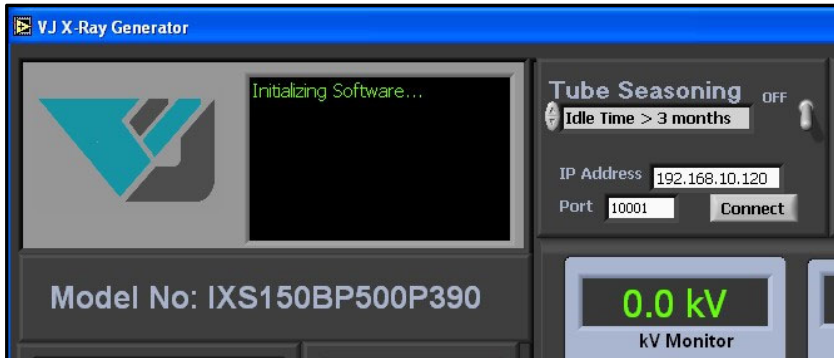


Fig 3-8 GUI Interface – Ethernet Connection

7. If you choose to connect the serial port, the GUI will open as shown in Fig 3-9 and begin automatically searching for the correct COM port used for serial communications among all available COM ports. When found and successfully connected it will display the COM port used in the upper left-hand corner just below **Initializing Software...**

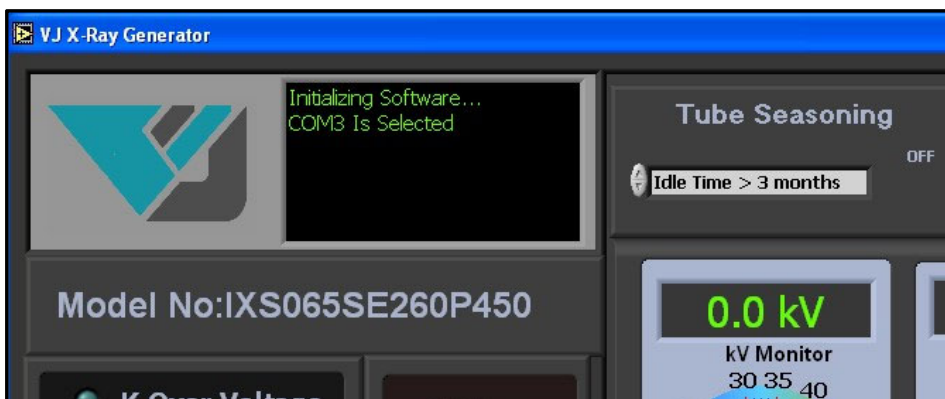


Fig 3-9 GUI Interface – Serial Connection

8. Notice the GUI is in constant communication with the IXS, for monitoring and fault detection. If communication is lost or it cannot connect the first time, a dialog will pop up as shown in Fig 3-10.
9. It will allow you to try to connect again or close the program. If you have trouble connecting, make sure the serial cable is properly setup in Windows and connected to the IXS's serial or ethernet connector.



Fig 3-10 Communications Failure Error Dialog

10. If you wish to close the GUI then click the **X** in the upper right-hand corner, then click **OK** in the pop-up window that says, "Do you want to close the Application?" as shown in Fig 3-11.

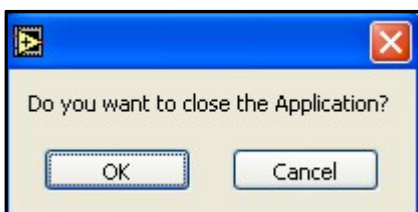


Fig 3-11 Close GUI Pop-up Window

3.2 Tube Seasoning

When operating the unit for the first time or the unit has been idled for excessive time, one needs to perform the warmup procedure. Select the suitable Idle time recipe from the drop-down control in the **Tube Seasoning** section, then click the knob to the right of it to flip it from **OFF** to **ON**, and finally press the **X-Ray On** button as shown in Fig 3-12. To stop press the **X-Ray On** button again to turn off the X-Rays, or it will automatically stop when the tube seasoning is done.

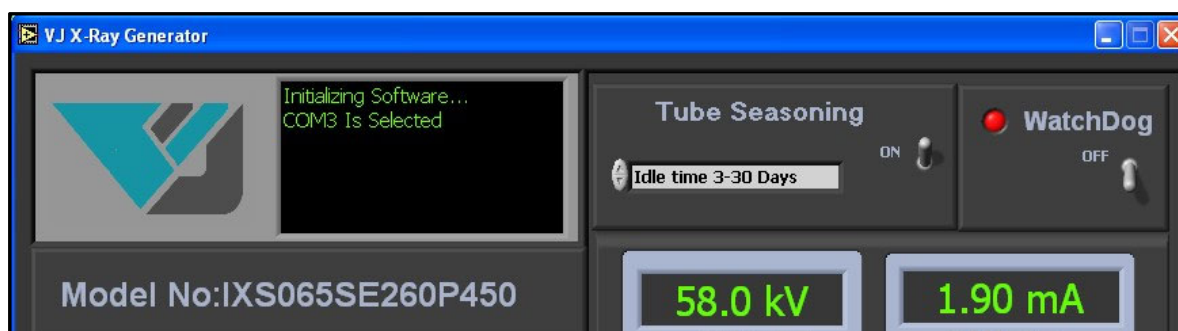


Fig 3-12 Seasoning

The below table shows a general seasoning schedule for the IXS products.

Time since last full kV use	Seasoning
Less than 3 days	No Seasoning required
Between 3 to 30 days	Short Seasoning (approx. 5 min in 10 Steps)
Between 1 to 3 months	Medium Seasoning (approx. 10 min in 10 Steps)
More than 3 months	Longest Seasoning (approx. 50 min in 10 Steps)

3.3 Operating the X-Ray

When the IXS completes the seasoning cycle, then normal operation can start.

Enter the desired kV value by entering a value above **kV Program** or by adjusting the knob above it.

Enter the desired mA value by entering a value above **mA Program** or by adjusting the knob above it.

Once both kV and mA are set to the desired values, then press the **X-Ray OFF** button to turn on the X-Rays as shown in Fig 3-13. Notice the windows above the programmed values show the output **kV Monitor** and **mA Monitor** and that the **X-RAY ON light** is lit above the **X-Ray ON** button, which is also lit. To stop press the **X-Ray ON** button and the X-Rays will turn off with the **X-RAY ON light** now off and button light too, and the **kV and mA Monitor** values will go back down to zero.

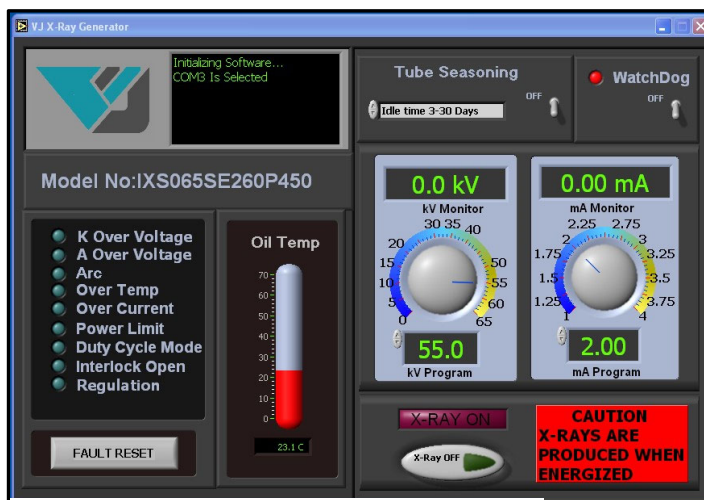


Fig 3-13 Enter kV and mA

3.4 X-Ray on-Time, Minimization and Pre-Warning

Some IXS units have special features such as Cumulative X-Ray On-Time, Minimization or Pre-Warning. This section describes the GUI interface such features.

3.4.1 X-Ray On-Time

The Cumulative X-Ray On-Time is a feature that keeps track of the total amount of time the X-Rays have

been on over the life of the unit. The time is expressed in hours and minutes, stored in the control box's nonvolatile memory, and displayed on the GUI in the upper-right hand corner as shown in Fig 3-14.



Fig 3-14 GUI with Cumulative X-Ray On-Time Feature

3.4.2 Minimization

The Minimization feature can only be activated when the X-Rays are on. It allows the X-Ray output to be minimized by setting mA and Filament DAC's to 0V by the X-Ray Controller while keeping the kV the same. This allows the X-Ray to be turned off and on quickly without having to wait for kV ramping. If the feature is present, then the GUI will have a knob that allows it to be activated or deactivated during normal X-ray operation as shown in Fig 3-15.



Fig 3-15 GUI with Minimization Feature

3.4.3 Pre-Warning

The Pre-Warning feature allows the X-rays start to be delayed by a certain number of seconds while a special warning light turns on. It is a safety feature that allows time for the operator to clear a certain area around the X-Ray system before they actually turn on. Please see section 2.3 for hardware wiring.

You can set the pre-warning period by pressing the **SET PTime** button (Fig 3.16) to open the dialog shown in Fig 3-17. In the **PreWarning** dialog enter the Pre-Warning value in the box, or use the knob, and press the **SET** button or press the **Cancel** button to cancel that value.

When pre-warning is activated, the "PRE-WARN" banner is highlighted in the GUI and turned off when the X-Rays are ready to turn on, as shown in Fig 3.16.



Fig 3-16 GUI with Pre-Warning Feature



Fig 3-17 Set Pre-Warning Timer

4.0 Fault Diagnostics

The IXS comes equipped with the technology to detect faults like arcing, over temperature, over current, over voltage. All these fault conditions are monitored in real time while the IXS is running. If a fault is detected, either the GUI and/or LED panel will present what type of fault. This section describes the faults, facilitates the diagnoses and how to clear the faults.

4.1 Fault Detection

Use below fault explanation to help further diagnostic root cause. Fig 4.1 show where various fault indications are located on the GUI.

1. **K Over Voltage:**
This will trigger when the Cathode side of the X-ray tube is over voltage. The trip point is set within 105% to 110% of max. kV. It will disable the X-ray output.
2. **A Over Voltage:**
This will trigger when the Anode side of the X-ray tube is over voltage. The trip point is set within 105% to 110% of max. kV. It will disable the X-ray output.
3. **Arc:**
When an Arc occurred, the arc fault LED will be ON. If 4 or more arcs occurs within 10 seconds, the arc fault signal will be latched. This will disable the X-ray output.
4. **Over Temp:**
Over temperature trip point is set within 57°C to 63°C. This will disable the X-ray output.
5. **Over Current:**
The over current trip point is set at +5% of maximum mA. It will disable the X-ray output when reach the trip point.
6. **Power Limit Fault:**
When the kV/mA feedback values exceed the maximum rated power limit. This will disable the X-ray output.
7. **Duty Cycle Mode (when equipped):**
This trigger when X-ray output power exceeded recommended constant power. This is NOT a fault.
8. **Interlock Open:**
This fault will trigger when the external interlock switch (section 2.3) is open.
9. **Regulation:**
When kV or mA output is out of regulation, the unit will disable the X-ray output.

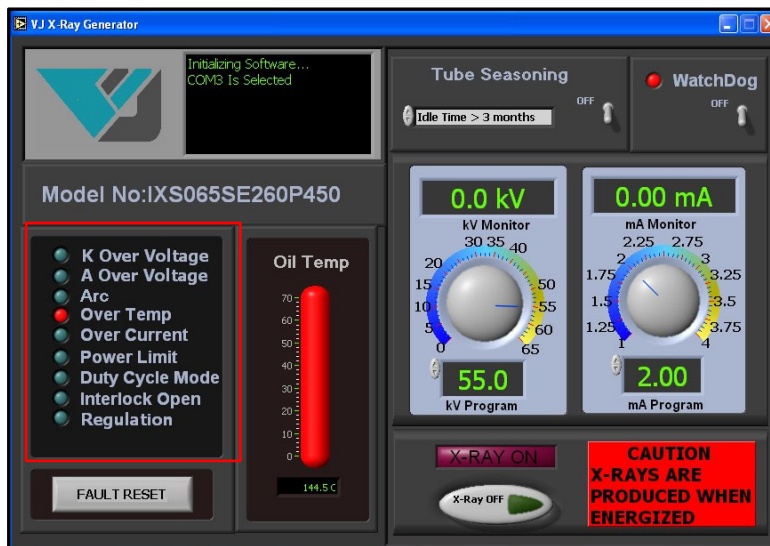


Fig 4-1 Faults

4.2 LED Indicators

In addition to the GUI fault indications, the control panel also has six LED indicators (Fig 4-2) that present same information. This is useful when VJX's GUI is not used to control the IXS.

POWER (Green)	Illuminated when Power is present
X-Ray On (Red)	Illuminated when Interlock is closed & HV is enabled
OV (Yellow)	Over Voltage fault
OT (Yellow)	Illuminated when oil or inverter temperature exceeds the set threshold
ARC (Yellow)	Arcing Fault
OC (Yellow)	Over Current Fault
OP (Yellow)	Overpower. This LED will illuminate when selected power exceeds the rated power

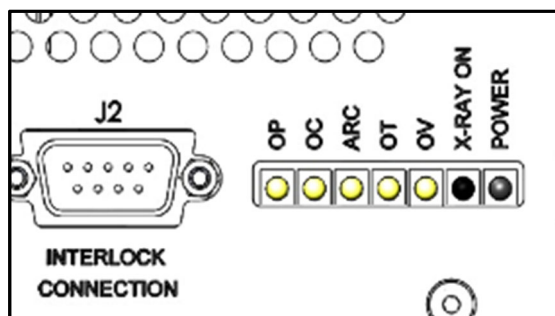


Fig 4-2 LED

4.3 Troubleshooting

If the IXS is not working as it should, has been dropped, damaged, please contact VJ-X-Ray technical support. Do not open the unit without explicit directions from VJX technical support (1-800-VJT-XRAY or vjxservice@vjt.com)

Problems	Possible Causes	Solutions
No input power / Green Power light off	No power	Check Input power
	Wrong power	Check Input voltage range
GUI No communication	No power	Make sure to apply power to the unit.
	Wrong customer GUI	Make sure to apply corresponding customer GUI.
	Computer access/software conflict	Get permission to use the computer or replace a computer.
	Cable poor connection	Make sure the Serial or Ethernet cable is securely connected, also check cable for damage.
	Ethernet communication priority	If both Serial and Ethernet are connected, only ethernet will work.
	PC Serial com port fail	For Serial connection, check PC com port, make sure it is enable.
	Wrong IP address setting	For Ethernet, make sure the IP address is set at 192.168.10.120 or see section 5.3
	USB to Serial cable converter	If USB to Serial Converter is used, make sure the device driver is updated.
	Interlock open	Close the interlock.
Cannot turn on X- Ray	Fault alarm protection	Any fault shows, unit needs to reset after any fault.
	Wrong customer GUI	Make sure correct GUI is used.
	Poor input power	Check the AC input power is stable when turn on X- Ray.
	Connector poor connection	Check all connectors securely connected. Re-plug them if needed.
K Over Voltage Fault	X-ray tube cathode side over voltage	Reset to clear. If fault repeat, contact VJX technical support.
A Over Voltage Fault	X-ray tube anode side over voltage	Reset to clear. If fault repeat, contact VJX technical support.

Problems	Possible Causes	Solutions
Over-Temp Fault	External (if equipped) Chiller Stop	Make sure the chiller is running.
	External Chiller (if equipped) lack of coolant	Add enough coolant for chiller.
	High ambient temperature	Make sure the ambient temperature is meet the IXS specification.
	Radiator clogged w/ dust	Clean the radiator with a hard brush or compress air
	Radiator fan not turning	Check the fan for dust and power. Does the fan blades spin?
Regulation Fault	No output current	Make sure the minimum kV program setting is matched with the specification.
	No output voltage	Make sure kV and mA program is setting correctly, interlock is closed and enable is on.
Arc Fault	X-Ray tube or External arcs	X-Ray tube arc: Need longer seasoning
		External arc: Multiple arcs occurred within 10 second, unit shuts down and cannot run further: Contact VJX technical support.
Power Limit Fault	The kV/mA exceed maximum allowed	Make sure kV/mA (section 3.3) does not exceed the specified power limit.
Watchdog Fault	Watchdog Fault	The fault is resulted from IXS to PC's communication link broken while the unit is running. Check Serial or Ethernet link.
Interlock Open Fault	Interlock connector wiring open	Check J2 connector wiring
	Interlock connector fail	Check cabinet interlock
Over Current Fault	Over current	Enable the X-Ray with low kV (10kV). If over current repeat, contact VJX-ray technical support.

5.0 Addendum

5.1 Regulatory Compliance

The IXS units are certified by Intertek Testing Services under:

- Safety Requirements for Electrical Equipment, Measurement Control and Laboratory Use Part 1: General Requirements (UL 61010-1 and IEC 61010-1)
- Electrical equipment for measurement, control and laboratory use - EMC requirements - Part 1: General requirements (EMC IEC 61326-1)

Conformity to CE Mark in reference to Low Voltage Directive (2014/35/EU) and Electromagnetic Compatibility Directive (2014/30/EU).

5.2 Win 10 Run as Administrator

If you have **Windows 10 or up**, you need to run the setup.exe and the GUI program in Administrator mode. To set the program as administrator mode:

1. Right-click the program and go to Properties.
2. Click on the Compatibility tab.
3. Check **Run as Administrator** checkbox. **Run as administrator** option for program.

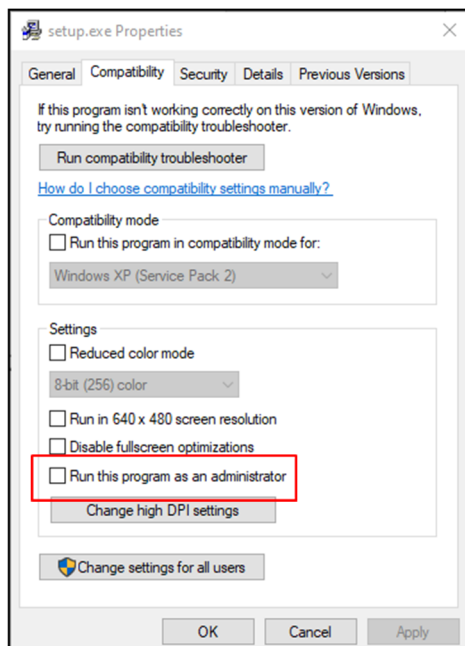


Fig 5-1 Administrator Mode

5.3 Ethernet IP Address Programming

1. Connect the PC directly to the Ethernet connection via a regular Ethernet cable.
2. Make sure the unit is powered up and launch the utility by double clicking on **USR-TCP232-T24**

V5.1.1.20.exe in the CD.

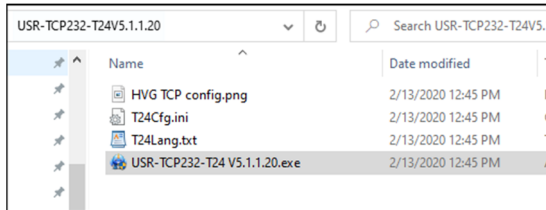


Fig 5-2 TCP232 program

- Then in the utility, click on [Search in LAN] button and make sure default settings match.

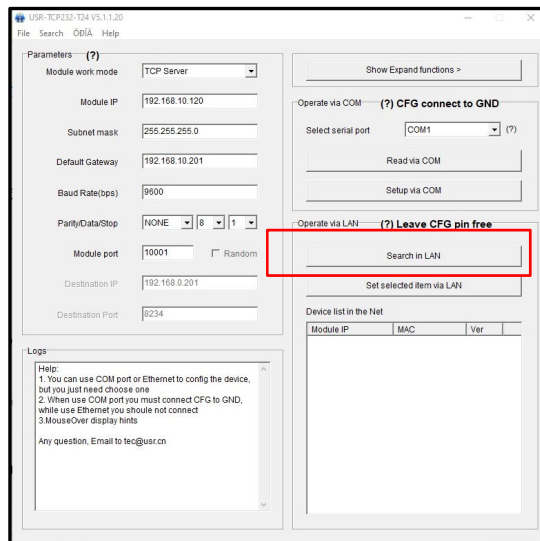


Fig 5-3 Run USR-TCP program

- The [Module work mode] in the [Parameters] section should be set to [TCP Server]. Don't change it.
- Change [Module IP], [Subnet Mask], and [Default Gateway] to the desired values.
- The [Baud Rate] should stay at [9600 bps].
- The [Parity/Data/Stop] should stay at [None/8/1].
- Change [Module Port] to desired value also for TCP port. Although it's best to leave it set to the default value of 10001.
- Finally, click on the [Set selected item via LAN] button.
- Also, you need to change the PC's IP settings to a static IP address with the same subnet mask and default gateway and with an IP addresses that's in the same subnet as the generator.

Note: Default Gateway is not important with a direct connection to the generator, but it is crucial through a router or gateway. When connect through a router or gateway, please make sure the default gateway is correct and that both are on the same subnet.

Revision History

Revision	Date	Remark
Rev 1	June 2012	Initial Release
Rev 2	June 2020	Add: - Add software operation - Add fault diagnostics - Addendum

End of Document